

Mechanics and physics of solids at micro- and nano-scales

Conference Agenda

Tuesday, 19th of June

Chimie ParisTech, Amphithéâtre Chaudron

8:30 to 9:00	Registration
9:00 to 9:20	Welcome Ioan R. Ionescu (ICACM) Dominique Vrel (LSPM) Christian Ricolleau (SEAM)
9:20 to 10:00	Damage and plastic deformation of crystals (I), Chair Oana Cazacu Alan Needleman <i>Shear band development in a planar single crystal under dynamic loading</i>
10:00 to 10:40	Lev Truskinovsky <i>Critical aspects of brittle fracture</i>
10:40 to 10:50	Posters session: Short Oral Presentations
10:50 to 11:20	Coffee Break/Posters
11:20 to 12:00	Mechanics of fibrous soft matter (I), Chair Catalin Picu Prashant Purohit <i>Compression of fibrin and CNT networks viewed as a phase transition</i>
12:00 to 12:40	Laurent Orgeas <i>Rheology of fiber suspension using 3D imaging and X-ray microtomography</i>
12:40 to 14:30	Lunch (Le Perraudin)
14:30 to 15:10	Dislocations and interfaces (I), Chair Sylvain Queyreau Sylvie Aubry <i>Dislocation Dynamics Simulations of Materials with Complex Physics</i>
15:10 to 15:50	Emmanuel Clouet <i>Atomic-Scale modeling of Screw Dislocation Mobility in Zr and Ti</i>
15:50 to 16:00	Posters session: Short Oral Presentations
16:00 to 16:30	Coffee Break/Posters
16:30 to 17:10	Dislocations and interfaces (II), Chair Emmanuel Clouet Stefanos Papanikolaou <i>Rare events and uncertainty quantification in crystal plasticity</i>
17:10 to 17:50	David Rodney/ Pierre-Antoine Geslin <i>Dislocation vibrations reveal the interplay between long-range elasticity and short-range line tension</i>
19:30	Welcome Party (Café Les Editeurs)

Wednesday, 20th of June

Chimie ParisTech, Amphithéâtre Chaudron

09:00 to 09:40	<i>Mechanics of fibrous soft matter (II), Chair Marc Durand</i> Catalin Picu <i>Structure and mechanics of self-organized fibrous materials with inter-fiber adhesion</i>
09:40 to 10:20	Jean-Marc Allain <i>Multi-scale mechanics of soft tissues</i>
10:20 to 10:50	Coffee Break
10:50 to 11:30	<i>Damage and plastic deformation of crystals (II), Chair Umut O. Salman</i> Turab Lookman <i>On yield in amorphous solids under periodic shear</i>
11:30 to 12:10	Oana Cazacu <i>New Constitutive Modeling of Plastic Anisotropy: from Single-crystal to Polycrystalline Response</i>
12:10 to 12:50	Yann Charles <i>Finite Element simulations of hydrogen-plasticity interactions in metallic polycrystals</i>
12:50 to 14:40	Lunch (<i>Restaurant Le Coupe Chou</i>)
14:40 to 15:20	<i>Nonlinear solids and wetting (I), Chair Laurent Limat</i> Anand Jagota <i>Response of a Surface with Extensional and Bending Elasticity</i>
15:20 to 16:00	François Lequeux <i>Heterogeneous mechanics of polymers in the glass transition regime, consequences and modeling</i>
16:00 to 16:10	Posters session: Short Oral Presentations
16:10 to 16:40	Coffee Break/Posters
16:40 to 17:20	<i>Nonlinear solids and wetting (II) Chair Francois Lequeux</i> Alexandre Danescu <i>Geometric incompatibilities and bending of ultra-thin films</i>
17:20 to 18:00	Benoît Roman <i>Surface effects on elastic structures</i>
20:00	Banquet (<i>Restaurant Bouillon Racine</i>)

Thursday, 21st of June

Chimie ParisTech, Amphithéâtre Chaudron

9:00 to 9:40	<i>Nonlinear solids and wetting (III)</i> , Chair Alexandre Danescu Laurent Limat <i>Mechanics of wetting on highly deformable gels</i>
9:40 to 10:20	Sebastien Neukirch <i>Liquid forces on elastic structures: capillarity prevails at small scales</i>
10:20 to 10:30	Posters session: Short Oral Presentations
10:30 to 11:00	Coffee Break-Posters
11:00 to 11:40	<i>Dislocations and interfaces (III)</i> , Chair Sylvie Aubry Aurélien Vattré <i>Interfaces in microstructure during phase transitions</i>
11:40 to 12:20	Oguz Umut Salman <i>Collective nucleation of dislocations accounting for lattice effects</i>
12:20 to 14:20	Lunch (<i>Restaurant La Méthode</i>)
14:20 to 15:00	<i>Mechanics of fibrous soft matter (III)</i> , Chair Jean-Marc Allain Stepan Lomov <i>Morphology and mechanical behaviour of fibrous materials as revealed via micro-computed tomography</i>
15:00 to 15:40	Marc Durand <i>Elastic networks with optimal mechanical properties</i>
15:40-16:10	Conclusions and Farewell Coffee

Addresses

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École nationale supérieure de chimie de Paris,
11, rue Pierre et Marie Curie, 75231 Paris.
<https://www.chimie-paristech.fr/>

Hôtel de Senlis,
7-9, rue Malebranche, 75005 Paris.
<https://paris-hotel-senlis.com/fr/>

Tuesday June 19th, Lunch.
Restaurant Perraudin,
157, Rue Saint-Jacques, 75005 Paris.
<http://www.restaurant-perraudin.com/>

Tuesday June 19th, 7:30 pm, Welcome reception.
Restaurant Les Editeurs,
4, carrefour de l'Odéon, 75006 Paris.
<http://www.lesediteurs.fr/>

Wednesday June 20th, Lunch
Restaurant Le Coupe-Chou,
9 & 11, rue de Lanneau, 75005 Paris.
<http://www.lecoupechou.com/>

Wednesday June 20th, 8 pm, Banquet.
Restaurant Bouillon Racine,
3 rue Racine 75006 Paris.
<http://www.bouillonracine.com/en/>

Thursday June 21st, Lunch
Restaurant La Méthode,
2 rue Descartes, 75005 Paris.
<http://www.restaurantlamethode.fr/>